RAVI SHANKAR

GANITA MANJARI

CHAPTER 5 – CIRCLES

EXERCISE 5.5

Question 1.
Find the length of the chord of a circle where the radius is 7 cm and perpendicular distance is 6 cm.

Solution

Let AB be the chord of the circle and O be its centre.

Let OM be the perpendicular distance from the centre to the chord.

Given,

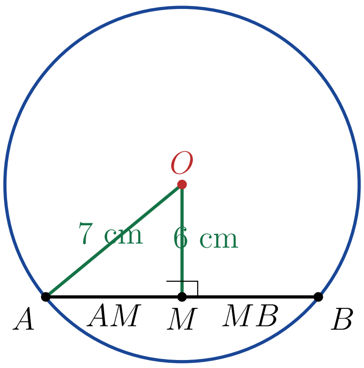
$$OA = 7 \text{ cm}$$

and

$$OM = 6 \text{ cm}.$$

Since the perpendicular from the centre of a circle to a chord bisects the chord,

$$AM = MB.$$



In right-angled triangle OMA, by Pythagoras theorem,

$$OA^2 = OM^2 + AM^2.$$

Substituting the given values,

$$7^2 = 6^2 + AM^2.$$

Therefore,

$$49 = 36 + AM^2,$$

$$AM^2 = 13.$$

Hence,

$$AM = \sqrt{13} \text{ cm}.$$

Since M is the midpoint of the chord,

$$AB = 2AM.$$

Therefore,

$$AB = 2\sqrt{13} \text{ cm}.$$

Answer: $\boxed{2\sqrt{13} \text{ cm}}$

Question 2.
Explain why the following statement is true:
If the perpendicular distance of a chord from the centre is d and the radius is r, then the chord length is

$$2\sqrt{r^2 - d^2}.$$

Solution

Let AB be a chord of a circle with centre O.

Let OM be the perpendicular distance from the centre to the chord.

Therefore,

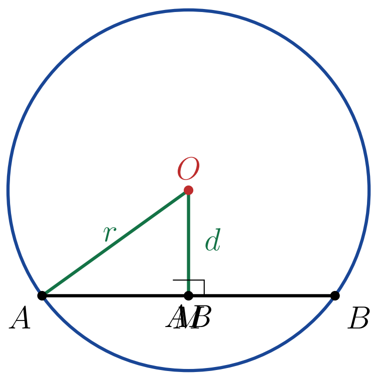
$$OM = d.$$

Let the radius of the circle be

$$OA = r.$$

Since the perpendicular from the centre to a chord bisects the chord,

$$AM = MB.$$



Consider the right-angled triangle OMA.

By Pythagoras theorem,

$$OA^2 = OM^2 + AM^2.$$

Since

$$OA = r$$

and

$$OM = d,$$

we get

$$r^2 = d^2 + AM^2.$$

Therefore,

$$AM^2 = r^2 - d^2.$$

Hence,

$$AM = \sqrt{r^2 - d^2}.$$

But M is the midpoint of the chord, so

$$AB = 2AM.$$

Therefore,

$$AB = 2\sqrt{r^2 - d^2}.$$

Thus, the length of the chord is

$$\boxed{\text{Chord length} = 2\sqrt{r^2 - d^2}}$$

Question 3.
In a circle, if the distance of chord AB from the centre is twice the distance of another chord CD from the centre, then can we conclude that $CD = 2AB$?
Give reasons for your answer.

Solution

No, we cannot conclude that

$$CD = 2AB.$$

Let the radius of the circle be r.

Suppose the perpendicular distance of chord AB from the centre is d.

Then the distance of chord CD from the centre is $2d$.

For a chord whose perpendicular distance from the centre is x, its length is

$$2\sqrt{r^2 - x^2}.$$

Therefore, for chord AB,

$$AB = 2\sqrt{r^2 - d^2}.$$

For chord CD,

$$CD = 2\sqrt{r^2 - (2d)^2}.$$

Hence,

$$CD = 2\sqrt{r^2 - 4d^2}.$$

Clearly,

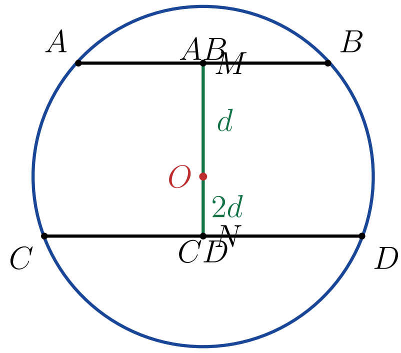
$$2\sqrt{r^2 - 4d^2} \neq 2\left(2\sqrt{r^2 - d^2}\right)$$

in general.

Therefore,

$$CD \neq 2AB$$

in general.



In fact, increasing the distance of a chord from the centre makes the chord shorter, not longer.

For example, let

$$r = 5 \text{ cm}, \qquad d = 2 \text{ cm}.$$

Then

$$AB = 2\sqrt{5^2 - 2^2}$$

$$= 2\sqrt{25 - 4}$$

$$= 2\sqrt{21} \text{ cm}.$$

The distance of CD is $2d = 4$ cm. Hence,

$$CD = 2\sqrt{5^2 - 4^2}$$

$$= 2\sqrt{25 - 16}$$

$$= 2(3)$$

$$= 6 \text{ cm}.$$

But

$$2AB = 4\sqrt{21} \text{ cm},$$

which is clearly not equal to 6 cm.

Therefore, the statement is false.

Answer: $\boxed{\text{No, } CD = 2AB \text{ cannot be concluded.}}$

IMPORTANT FORMULA

If r is the radius and d is the perpendicular distance of a chord from the centre, then

$$\boxed{\text{Chord length} = 2\sqrt{r^2 - d^2}}$$

Also,

$$\boxed{\text{Greater distance from centre} \implies \text{Shorter chord}}$$